Why a Linear Seasonal Baseline is Accurate at Every Bucketing Granularity

Traffic Forecasting Project

June 2026

Abstract

The calendar-adjusted seasonal baseline – a recency-weighted average of the same hour-of-week, times a calendar factor – is empirically accurate whether applied to a single sensor at hourly resolution, a region total, or a weekly aggregate. We explain why with two short results. *Equivariance:* the baseline is a linear operator and therefore commutes with any linear aggregation, so the same model is automatically self-consistent at every granularity (forecasting then aggregating equals aggregating then forecasting). *Signal-to-noise scaling:* the forecastable part is a low-rank, shared seasonal signal while the residual is high-rank idiosyncratic noise, so aggregation amplifies signal coherently ($\propto n$) and averages noise incoherently ($\propto \sqrt{n}$), making the same model at least as accurate – usually more – as the buckets coarsen.

1 Setup

Let $y_{i,t}$ be the series for unit $i \in \{1, \ldots, N\}$ (e.g. a sensor) at hour t, with weekly period $P = 168$. The *seasonal baseline* is the linear filter

$$\hat{y}_{i,t} \;=\; (By)_{i,t} \;=\; \sum_{k=1}^{K} w_k\, y_{i,\,t-Pk}, \qquad w_k \geq 0, \quad \sum_{k=1}^{K} w_k = 1, \tag{1}$$

i.e. a (possibly decaying) average over the same hour-of-week in the previous K weeks. The flat 4-week rule is $K = 4$, $w_k = \frac{1}{4}$; the EWMA rule is $w_k \propto \alpha^{k-1}$. The crucial structural facts about B are that it is **linear**, **time-invariant**, and uses **the same weights for every unit** i.

A *granularity* is produced by a linear *aggregation operator* A that maps the fine series to a coarser one by summing over blocks of indices:

$$\text{spatial:} \quad (A^{\text{sp}}y)_{R,t} = \sum_{i \in R} y_{i,t} \quad \text{(sum over a set of units } R\text{)}, \tag{2}$$

$$\text{temporal:} \quad (A^{\text{tw}}y)_{i,w} = \sum_{h=0}^{P-1} y_{i,\,Pw+h} \quad \text{(sum the 168 hours of week } w\text{)}. \tag{3}$$

(Means are the same up to a constant, so the argument is identical for averages.)

We decompose the series into a predictable seasonal/level signal and zero-mean noise,

$$y_{i,t} \;=\; \mu_{i,t} \;+\; \varepsilon_{i,t}, \qquad \mu_{i,t} = \ell_{i,t}\, s_{i,\phi(t)}, \quad \mathbb{E}[\varepsilon_{i,t}] = 0, \tag{4}$$

where $\phi(t) = t \bmod P$ is the hour-of-week, $s_{i,\cdot}$ is a (slowly varying) weekly *shape*, $\ell_{i,t}$ a slow *level*, and ε is the idiosyncratic, incident-driven residual. The baseline B is a near-unbiased, low-variance estimator of μ (it averages recent same-phase values); its error is dominated by the irreducible noise ε.

2 Equivariance: one model at every granularity

Proposition 1 (Aggregation equivariance)**.** *For spatial aggregation, B commutes with A^{sp} exactly:*

$$B\,A^{\mathrm{sp}} \;=\; A^{\mathrm{sp}}\,B.$$

For whole-week temporal aggregation, the hourly baseline summed over a week equals the same-form *weekly baseline applied to the weekly totals:*

$$A^{\mathrm{tw}}B \;=\; B'\,A^{\mathrm{tw}},$$

where B' has identical weights w_k but at weekly lags.

Proof. Spatial, using linearity and that w_k does not depend on i:

$$\begin{aligned}(BA^{\mathrm{sp}}y)_{R,t} &= \sum_k w_k (A^{\mathrm{sp}}y)_{R,t-Pk} = \sum_k w_k \sum_{i\in R} y_{i,t-Pk} \\ &= \sum_{i\in R}\sum_k w_k\, y_{i,t-Pk} = (A^{\mathrm{sp}}By)_{R,t}.\end{aligned}$$

Temporal, using that the lags Pk are whole multiples of the aggregation block P:

$$\begin{aligned}(A^{\mathrm{tw}}By)_{i,w} &= \sum_{h=0}^{P-1}\sum_k w_k\, y_{i,\,Pw+h-Pk} = \sum_k w_k \sum_{h=0}^{P-1} y_{i,\,P(w-k)+h} \\ &= \sum_k w_k\,(A^{\mathrm{tw}}y)_{i,\,w-k} = (B'A^{\mathrm{tw}}y)_{i,w}.\end{aligned}$$

□

Corollary 1 (Self-coherence)**.** *The fine-level forecasts sum exactly to the coarse-level forecast: $\sum_{i\in R}\hat{y}_{i,t} = \widehat{(A^{\mathrm{sp}}y)}_{R,t}$, and likewise for time. Hence* a single model is simultaneously the model at every aggregation level*: one need not refit a different form per granularity, and forecasts made at different resolutions are automatically consistent.*

Remark 1 (The multiplicative calendar factor)*.* The calendar-adjusted forecast is $\hat{y}^{\mathrm{adj}}_{i,t} = f_{i,t}\,(By)_{i,t}$. Aggregating, $\sum_{i\in R} f_{i,t}(By)_{i,t} = f_{R,t}\sum_{i\in R}(By)_{i,t}$ holds *exactly* when the factor is uniform across the aggregated units, $f_{i,t} \equiv f_{R,t}$. A *pooled* factor (one value per (holiday, hour-of-day), shared across units) satisfies this, so the calendar correction is coherent too; a heterogeneous per-unit factor makes the aggregate factor a baseline-weighted average of the unit factors, so equivariance becomes approximate. The linear backbone (1) is always exactly coherent.

Remark 2 (Same weights)*.* Equivariance is exact only if the *same* weights w_k are used at each level. Re-tuning the decay per granularity breaks it slightly; in practice a shared α keeps the hierarchy coherent.

3 Signal-to-noise improves under aggregation

Equivariance says the same model *form* is correct everywhere; it does not by itself say the model is *accurate* everywhere. Accuracy follows from how signal and noise scale under aggregation.

Write the aggregate of n units (a region, or one unit's weekly total over $n = P$ hours) and split it via (4) into an aggregate signal and an aggregate noise. Two facts distinguish them:

- **The signal is shared (coherent).** Every unit carries the *same* weekly rhythm $s_{\phi(t)}$, so the seasonal components add in phase: the aggregate signal grows like $n\,\bar{\mu}$, and its variation across t grows like n times the per-unit variation.

- **The noise is idiosyncratic (incoherent).** Incidents are local. With per-unit noise variance σ^2 and pairwise correlation ρ,

$$\mathrm{Var}\Big(\textstyle\sum_{i=1}^{n}\varepsilon_{i,t}\Big) = n\sigma^2 + n(n-1)\rho\,\sigma^2 = n\sigma^2\big[1+(n-1)\rho\big]. \tag{5}$$

Proposition 2 (SNR scaling)**.** *Let the aggregate signal scale as n and the aggregate noise standard deviation be $\sigma\sqrt{n[1+(n-1)\rho]}$ from (5). Then the* relative *forecast error (noise-to-signal) of the baseline obeys*

$$\frac{\textit{aggregate noise s.d.}}{\textit{aggregate signal}} \sim \frac{\sigma}{\bar{\mu}}\sqrt{\frac{1+(n-1)\rho}{n}} \xrightarrow[n\to\infty]{} \begin{cases} 0, & \rho = 0 \ (\textit{idiosyncratic noise}),\\ \dfrac{\sigma}{\bar{\mu}}\sqrt{\rho}, & \rho > 0 \ (\textit{common component}).\end{cases}$$

Equivalently, the unexplained-variance fraction $1-R^2$ shrinks monotonically as the buckets coarsen, to 0 if the residual noise is purely idiosyncratic and to a floor $\propto \rho$ if it has a shared component.

Proof. Divide the aggregate noise s.d. $\sigma\sqrt{n[1+(n-1)\rho]}$ by the aggregate signal $\sim n\bar{\mu}$ to get $\frac{\sigma}{\bar{\mu}}\sqrt{(1+(n-1)\rho)/n}$. For $\rho = 0$ this is $\frac{\sigma}{\bar{\mu}}n^{-1/2} \to 0$; for $\rho > 0$, $(1+(n-1)\rho)/n \to \rho$, giving the stated floor. Monotonicity in n follows since $(1+(n-1)\rho)/n$ is decreasing for $\rho < 1$. □

So coarser granularity carries a higher signal-to-noise ratio: the same seasonal model – which captures μ and leaves ε – explains a larger fraction of variance. (Empirically the weekly R^2 exceeded the hourly R^2; part of that is this genuine noise-averaging, and part is the larger seasonal variance inflating the R^2 denominator – the former is the real effect.)

4 Synthesis

The two results compose. Using the factor structure $\mu_{i,t} = \ell_{i,t}s_{i,\phi(t)}$:

- the signal μ is **low-rank and shared** – a common deterministic seasonal pattern, present at *every* scale (self-similar: a weekly cycle within a week, an annual cycle across weekly totals);

- the noise ε is **high-rank and idiosyncratic**.

The baseline B is a near-unbiased, low-variance estimator of μ at any scale. Aggregation then does two complementary things: by Proposition 1 it *preserves the model's form and coherence*, and by Proposition 2 it *amplifies the shared signal while averaging out the idiosyncratic noise.* Equivariance gives consistency across granularities; SNR scaling gives (weakly) increasing accuracy. Together:

> *one simple, correctly-specified linear seasonal model is automatically self-consistent and accurate at every bucketing granularity.*

Caveats, all orthogonal to granularity. Exact equivariance needs the same weights at each level and (for the multiplicative factor) a pooled, uniform calendar factor. Correlated residual noise ($\rho > 0$) sets a floor on the SNR gain. And the *bias* properties are scale-independent: the baseline's level estimate goes stale as the forecast horizon grows, and its structural assumption breaks at regime shifts (e.g. a demand shock) – these inflate error at *all* granularities, not at one.